Chapter

Analytical Chemistry

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1. Aqueous solution of cupric chloride forms a deep blue solution on addition of MAIN 2021

(a) drop wise sodium hydroxide
(b) excess sodium hydroxide
(c) drop wise ammonium hydroxide
(d) excess ammonium hydroxide

2. If the empirical mass of the formula PQ2 is 10 and the relative molecular mass is 30, then the molecular formula will be COMP 2021

(a) PQ_2
(b) $P_3 Q_2$
(c) $P_6 Q_3$
(d) $P_3 Q_6$

3. If the empirical formula of a compound is CH and its vapour density is 13, then its molecular formula will be (At. wt. C = 12 , H = 1) COMP 2021

(a) CH
(b) $C_2 H_2$
(c) $C_4 H_4$
(d) $C_3 H_3$

4. Amphoteric oxides and hydroxides react with an .... , ... to give salt and water only. COMP 2021

(a) Acid, Alkali
(b) Base, Alkali
(c) Acid, Base
(d) None of these

5. Salts of which elements are generally coloured : SQP 2023

(a) Inner-transition
(b) Normal
(c) Lanthanide's
(d) Transition

6. An example of weak alkali solution is .... COMP 2021

(a) Sodium hydroxide
(b) Nitrogen dioxide
(c) Potassium hydroxide
(d) Ammonium hydroxide

7. Sodium zincate and water is obtained on reaction of .... with concentrated caustic soda. MAIN 2023

(a) Lead
(b) Zinc
(c) Sodium
(d) Zinc hydroxide

8. Which one of the following salt solutions on reaction with excess of ammonium hydroxide solution gives a deep blue solution ? COMP 2023

(a) $CuSO_4$ (aq)
(b) $FeCl_3$ (aq)
(c) Al_2 (SO_4) (aq)
(d) $ZnSO_4$ (aq)

9. A yellow monoxide that dissolves in hot and concentrated alkali.

MAIN 2021

(a) ZnO
(b) CuO
(c) AgO
(d) PbO

10. What is the reaction of freshly precipitated aluminium hydroxide with caustic soda solution? Give equation. MAIN 2023

11. Give two examples of amphoteric hydroxides. SQP 2021

12. Name the reagent used to distinguish zinc nitrate solution from magnesium nitrate solution. COMP 2022

13. Choose from the following list of substances, as to what matches the description given below :
(Acetylene gas, aqua fortis, coke, brass, barium chloride, bronze, platinum).
An aqueous salt solution used for testing sulphate radical. MAIN 2021

14. Identify only the anion present in the following compound: MAIN 2024

(a) The compound on heating produces a colourless gas which turns lime water milky and has no effect on acidified potassium dichromate solution.
(b) The solution of the compound which on treating with concentrated sulphuric acid and freshly prepared ferrous sulphate solution produces a brown ring.

15. Rita was given an unknown salt for identification. She prepared a solution of the salt and divided it into two parts.

1. To the first part of the salt solution, she added a few drops of ammonium hydroxide and obtained a reddish-brown precipitate.
2. To the second part of the salt solution, she added a few drops of silver nitrate solution and obtained a white precipitate.

Name:
(a) the cation present and
(b) the anion present in the salt given for identification. MAIN 2024

16. What are the colour of the precipitates when ammonium hydroxide is added to the following solutions? (a) Iron(II) chloride (b) Iron(III) chloride (c) Lead nitrate (d) Zinc nitrate (e) Copper nitrate

17. Rotten egg smell is due to the liberation of:

(a) HCl gas
(b) H2 S gas

(c) Cl_2 gas
(d) SO_2 gas

18. The colour of the precipitate formed after the addition of a small amount of sodium hydroxide solution to an aqueous solution of ferric chloride is

COMP 2021

(a) gelatinous white
(b) pale blue
(c) reddish brown
(d) dirty green

Solution

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1. (d) excess ammonium hydroxide

When **excess ammonium hydroxide (NH_4OH)** is added to **cupric chloride solution**, it forms a **deep blue complex** called **tetraammine copper(II) complex.**

$Cu^{2+} + 4NH_3 \rightarrow [Cu(NH_3)4]^{2+}$

This complex gives the **deep blue colour**.

2. (d) P_3Q_6

$$n = \frac{\text{Molecular mass}}{\text{Empirical formula mass}}$$
$$= \frac{30}{10} = 3$$

$$\text{Molecular formula} = n \times \text{Empirical formula}$$
$$= 3 \times (PQ_2) = P_3Q_6$$

3. (b) C_2H_2

$$n = \frac{2 \times \text{Vapour density}}{\text{Empiorical formula wt.}}$$
$$= \frac{2 \times 13}{13} = 2$$

$$\text{Molecular formula} = n \times \text{Empirical formula}$$
$$= 2 \times \text{CH} = C_2H_2$$

4. (a) Acid, Alkali

Amphoteric oxides and hydroxides react with both acids and alkalis to form salt + water. So the correct pair is: ✓ Acid, Alkali

5. (d) Transition

Transition element salts have one or more unpaired d-electrons.
When light falls on these salts, the unpaired electrons jump to higher energy levels and absorb certain wavelengths of light.
The remaining wavelengths are reflected, giving the salts their distinctive colours.

6. (d) Ammonium hydroxide

Ammonium hydroxide is a **weak alkali** because its **OH^- ions do not fully dissociate** in water.
In contrast, strong alkalis like NaOH and KOH dissociate completely, giving a stronger basic solution.

7. (b) Zinc

When **zinc (Zn)** reacts with **hot concentrated sodium hydroxide (NaOH)**, it forms **sodium zincate ($Na_2[Zn(OH)_4]$)** and **hydrogen gas (H_2)**:

$$Zn + 2NaOH + 2H_2O \rightarrow Na_2[Zn(OH)_4] + H_2 \uparrow$$

This reaction occurs because zinc is **amphoteric**, meaning it reacts with both acids and strong bases.

8.

$$\mathrm{Cu_2+(aq) + 2NH_3(aq) + 3H_2O(l) \longrightarrow Cu(OH)_2(s) + 2NH_4^+(aq)}$$

The precipitate dissolves in excess ammonia to form a dark blue complex ion.

9. (d) PbO.

- **PbO (Lead(II) oxide)** is a yellow-colored monoxide.
- It is **amphoteric**, so it **dissolves in hot and concentrated alkali** to form plumbate ions:

$PbO+2NaOH \rightarrow Na_2PbO_2+H_2O$

- Other oxides like ZnO, CuO, AgO do not match both the yellow color and solubility property in hot alkali.

10.

$$Al(OH)_3 + NaOH \xrightarrow{\Delta} \underset{\substack{\text{Sodium} \\ \text{meta-aluminate} \\ \text{(Soluble)}}}{NaAlO_2} + 2H_2O$$

11. 1. Aluminium hydroxide, $Al(OH)_3$

2. Zinc hydroxide, $Zn(OH)_2$

- Amphoteric hydroxides can react with both **acids** and **bases** to form salts and water.

12. Ammonium hydroxide (NH_4OH) or Sodium hydroxide (NaOH).

13. Barium chloride.

14. The anions present are:

(a) **Carbonate (CO_3^{2-})** – indicated by evolution of CO_2 gas on heating, which turns lime water milky and does not affect acidified potassium dichromate.

(b) **Nitrate (NO_3^-)** – indicated by formation of a brown ring with concentrated H_2SO_4 and freshly prepared ferrous sulphate solution (Brown Ring Test).

15. The cation and anion present in the salt are:

(a) **Cation:** Fe^{3+} (from Ferric ion) – indicated by the reddish-brown precipitate with ammonium hydroxide.
(b) **Anion:** Cl^- (Chloride ion) – indicated by the formation of white precipitate with silver nitrate (AgCl).

16. (a) **Iron(II) chloride ($FeCl_2$)** – Dirty green precipitate, which slowly turns reddish-brown on standing.
(b) **Iron(III) chloride ($FeCl_3$)** – Reddish-brown precipitate.
(c) **Lead nitrate ($Pb(NO_3)_2$)** – White precipitate.
(d) **Zinc nitrate ($Zn(NO_3)_2$)** – White precipitate.
(e) **Copper nitrate ($Cu(NO_3)_2$)** – Bluish-white precipitate.

17. (b) H_2S gas.

The rotten egg smell is due to the liberation of **hydrogen sulphide (H_2S) gas**.

18. (c) reddish brown.

When a small amount of **sodium hydroxide (NaOH)** is added to **ferric chloride ($FeCl_3$)** solution, a **reddish-brown precipitate** of **ferric hydroxide, $Fe(OH)_3$** is formed.